



Technical Solution (TS)

TS Overview

Required PA Information

Intent

Designs and builds solutions that meet requirements.

Value

Provides a cost-effective design and solution that meets customer requirements and reduces rework.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

TS 1.1 Build a solution to meet requirements.

Level 2

TS 2.1 Design and build a solution to meet requirements.

TS 2.2 Evaluate the design and address identified issues.

TS 2.3 Provide guidance on use of the solution.

Level 3

TS 3.1 Develop criteria for design decisions.

TS 3.2 Develop alternative solutions for selected components.

TS 3.3 Perform a build, buy, or reuse analysis.

TS 3.4 Select solutions based on design criteria.

TS 3.5 Develop, keep updated, and use information needed to implement the design.

TS 3.6 Design solution interfaces or connections using established criteria.

Additional PA Explanatory Information

Activities for designing and building solutions can be applied:

- To products or product components
- To services, service systems, and service components

- At any level of the product or service architecture

Design and build solutions that meet customer, functional, and quality requirements by:

- Developing, evaluating, and selecting cost-effective design solutions. These selected design solutions can be called design approaches, design concepts, or preliminary designs.
- Developing designs detailed enough to support implementation of the selected design solutions.
- Implementing the designs as a product, service, or component.

Related Practice Areas

Product Integration (PI)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Agile teams build designs incrementally, “emergent design” after developing functionality during each Sprint. Emergent design can introduce risk when developing critical, complex, or large systems since design defects introduced early can be expensive to correct later.

It is typical for agile teams to demonstrate less definition, clarity, and recording of designs as compared to more traditional software development teams. Extensive use of white boards, cameras, and other temporary mediums are common among agile teams for recording designs.

Design activities provide a foundation to ensure that the design is developed, usually incrementally, prior to implementation, and the results are recorded to:

- Identify minimal viable product
- Efficiently share technical information with stakeholders
- Mitigate technical risks
- Peer review to find defects early
- Support maintenance

[Table TS-1: Design Activities in an Agile Project](#) shows where design activities are addressed in a typical agile project.

Table TS-1: Design Activities in an Agile Project

Agile Activities	Purpose
Release planning	Obtain earlier and more complete understanding of the end solution, completion risks, and appropriate sequencing.
Backlog grooming/review	Allocate requirements to design components to be developed for known user stories during this Sprint. This helps to identify additional requirements that may have been missed.

Agile Activities	Purpose
Sprint planning	Gain broader understanding for context of designs and interfaces or connections for the upcoming Sprint.
Sprint execution	Implement the design.
Sprint review	Determine what has been accomplished during the Sprint.
Sprint retrospective	Identify what design components remain.

Time is allocated to each Sprint to perform design practices. Design documentation can be in the form of a picture and bulleted design notes stored in the same tools used to store user stories or epics and other project data.

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Establish and keep updated architecture standards and criteria which address data considerations, e.g., data representations, data lineage, data security, data sovereignty, data access, data privacy, and data provisioning. Three key concepts to address within architecture standards include:

- Data representation, e.g., business terms, logical, physical, XML, modeling standards, model management, data resiliency
- Data access, e.g., common data services, applicable information exchange standards, standard methods for point-to-point data transit and bulk data movement, data integration standards for safety and emergencies, data security, data privacy
- Data distribution, e.g., internal and external data provisioning such as distribution control and management, data scalability, requesting and approving access, access restrictions, distribution models for push and pull, publish and subscribe, ownership and authority, regulatory authority and audit

Evaluate design decisions based on architecture standards, e.g., platform, technology, tools, to ensure resulting solutions meet business needs for managing data.

DevSecOps

Context Tag:	DevSecOps
Context:	DevSecOps is a mindset, a culture, and set of practices that fosters close cooperation between development, operations, and security to plan, develop, test, deploy, release, and keep updated a secure solution.

There is no standard DevSecOps pipeline; the design and implementation of a DevSecOps pipeline depends on the development group's technology stack, agile development processes, team skills and experience, and available resources. Due to the complexities involved, initial environment configuration can take multiple iterations to automate the development. Common pipeline technology solutions include Microsoft Azure and Amazon Web Services (AWS). DevSecOps teams must be trained and have knowledge of both development, operations, and security including secure coding, infrastructure management, system administration, and DevSecOps toolchains.

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

Consider safety concerns, constraints, hazards, and events throughout all aspects of product or solution design. Address safety needs like any other requirement or criteria in the design of the entire solution, e.g., components, interfaces, connections, and integration.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Most aspects of security are addressed multiple times within a solution design, but software design and development in particular can introduce many security vulnerabilities. However, few software design methodologies or lifecycles explicitly address software security in detail, so secure software design practices must be added to each element of the design to ensure that the software being developed is well-secured. For more information, refer to NIST Special Publication 800-218, Secure Software Development Framework.

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

It is important to remember that in some simple service systems the components are just the people and the processes they perform.

Service system development focuses on the following activities:

- Collecting, coordinating, analyzing, validating, and allocating stakeholder requirements for service systems
- Evaluating and selecting from alternative service system solutions that meet requirements
- Designing, building, or composing (as needed), integrating, and recording service systems that meet requirements

- Verifying and validating service systems to confirm satisfaction with intended requirements and satisfaction with customer and end user expectations during actual service delivery

The service organization chooses to develop the service system from a wide range of options, from internal development to outsourcing to commercial product integration. Most service organizations engage a development team to build a service system. Choose the development method(s) based on the requirements to fulfill and the service system components to develop. When service system components comprise product components, develop requirements, as well as operational concepts and scenarios for the system, that consider the objectives of the service system, business needs, customer needs, user needs, and the needs of other affected stakeholders.

Level 1

TS 1.1

Required Practice Information

Practice Statement

Build a solution to meet requirements.

Value

Provides the customer with a solution that implements the requirements and reduces the cost of rework.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Build a solution.	Examples of building a solution include: • Software is coded • Data are recorded • Services are provided • Parts are fabricated • Manufacturing processes are put into operation • Facilities are constructed • Materials are produced

Example Work Products

Example Work Products	Further Explanation
Product or service	

Level 2

TS 2.1

Required Practice Information

Practice Statement

Design and build a solution to meet requirements.

Value

Provides a structure to guide the implementation of a cost-effective solution that meets requirements and avoids rework.

Additional Required Information

The design describes solution structure, interfaces or connections, and functionality.

Explanatory Practice Information

Additional Explanatory Information

The two main types of design are preliminary design and detailed design.

The preliminary design defines the solution and the architecture.

The detailed design describes solution component structure and functionality. The design is detailed enough to allow the solution component to be implemented, fabricated, or acquired. The level of detail required is typically set in the architecture's standards and design rules.

The activities involved in these two designs may be iterative and overlap.

Example Activities

Example Activities	Further Explanation
Define the architecture.	May include: • Developing the structural relationships and interface or connection rules between elements • Developing the structure that will support and integrate the solution components needed to meet the requirements • Identifying major internal and external interfaces or connections • Defining component behavior and interaction • Recording definitions using an architecture description language • Developing infrastructure capabilities and services • Developing solution component templates, classes, or frameworks • Developing design rules and authority for making decisions

Example Activities	Further Explanation
	• Defining a process or thread model • Identifying major reuse approaches and sources • Ensuring traceability to requirements
Identify, develop, or acquire effective design methods or tools for the solution.	Examples of effective design techniques and methods include: • Prototypes and associated design lessons • Structural models • Object-oriented design • Essential systems analysis • Entity relationship models • Design reuse • Design patterns
Evaluate commercial off-the-shelf (COTS) products.	Care in evaluating and selecting COTS products and the supplier can be critical to the project if the COTS product: • Is a significant part of the project or solution • Represents significant risk Aspects to consider in the selection decision include proprietary issues and the availability of the products.
Develop a preliminary design.	The preliminary design is the foundation for the solution and the architecture, and may include: • Architectural styles and patterns • Component identification • System states and modes • Major component interfaces or connections • External product interfaces or connections • Algorithms to be employed • Data definition
Develop a detailed design.	May include: • Finalizing the architecture • Completing component and interfaces or connection descriptions • Optimizing design • Selecting legacy or commercial components • Verifying and validating requirements
Track requirements against design to ensure they are satisfied.	As the design matures, track the requirements assigned to lower-level solution components and verify that those requirements are satisfied.
Build the solution.	Includes the allocation, refinement, and verification of each product component. It also involves the coordination between the various product component

Example Activities	Further Explanation
	development efforts. Steps to build the solution may include: • Using effective methods to build the solution. For example: ○ Structured programming ○ Automatic code generation ○ Software code reuse ○ Fabrication methods ○ Computer aided design drawing ○ Simulation • Adhering to applicable standards. For example: ○ Language standards ○ Drawing requirements ○ Manufactured parts ○ Process and quality standards ○ Construction regulations ○ Legal regulations • Adhering to applicable criteria. For example: ○ Modularity ○ Clarity ○ Scalability ○ Reliability ○ Safety ○ Maintainability

Example Work Products

Example Work Products	Further Explanation
Architecture	The architecture defines the design structure needed to meet the requirements. Architectures should: • Adhere to design standards and best practices • Provide a foundation for developing solution components and interfaces or connections • Include input from operational concepts and scenarios • Be traceable to requirements
Component design	The design provides a specification of the solution, defining how functional interface or connection quality requirements will be met, and includes traceability to requirements.
Completed solution	

Related Practice Areas

Decision Analysis and Resolution (DAR)

Requirements Development and Management (RDM)

Context Specific

Safety

Context Tag:	CMMI-SAF
Context:	Use processes to incorporate safety considerations as an integral part of the solution, work, project, and organization.

If possible, eliminate identified potential hazards through design selection/changes. Leverage the established functional safety approach for the safety in order of precedence, and consider:

- Incorporation of safety devices: if unable to eliminate the hazard through design selection, reduce the safety risk to an acceptable level using protective safety features or devices
- Use of warning devices: if safety devices do not adequately lower the safety risk of the hazard, include a detection and warning system to alert personnel to the particular hazard
- Use of procedures and training: when it is not possible to eliminate hazards through design selection or to reduce the associated risk to an acceptable level with safety and warning devices, incorporate special procedures and training; procedures may include the use of personal protective equipment
- Incorporation of fail-safes and system redundancies to mitigate safety risks

For hazards assigned catastrophic or critical severity categories, avoid using warning, caution, or other written advisory as the only risk reduction method. In general, design changes can either eliminate or reduce the severity and probability of a hazard. The use of procedures and training can only reduce the probability of occurrence. This may still be effective in reducing the hazard risk but cannot eliminate the risk.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Consider security when developing design solutions, including design approaches, design concepts, and preliminary designs. Apply security design principles, e.g., Principle of Least Privilege (PoLP) and definition of trust boundaries, when developing solution designs. Evaluate design solutions and technologies against security requirements.

Consider security risk assessments and review external agency threat lists to support the decision-making process when designing solutions. Evaluate threats and vulnerabilities for each solution alternative and use them in the decision process. Baseline and control all these technologies and decisions in a secure environment.

TS 2.2

Required Practice Information

Practice Statement

Evaluate the design and address identified issues.

Value

Reduces cost by minimizing defects and ensuring that the solution meets requirements.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Verify that the design meets requirements.

Operational concepts and scenarios can be used during design reviews to evaluate how well the design meets its intended purpose.

Example Activities

Example Activities	Further Explanation
Determine what types of reviews to perform.	To evaluate the design, complete a technical review of the solution and identify deficiencies or potential improvements. Types of evaluations include: • Structured design walkthroughs • Inspections
Identify review participants.	Participants can include: • Authors • Technical team members • Project managers • Subject Matter Experts (SMEs)
Send draft designs to reviewers.	Send these in advance so participants have enough time to review them.
Conduct a technical review.	Develop a review checklist to be available during the review. Use technical reviews to: • Present the design to affected stakeholders to develop a common understanding and gather input regarding the solution being considered • Decide the validity of the solution • Identify issues and concerns with the solution • Verify that the technical concepts are:

Example Activities	Further Explanation
	○ Correct ○ Consistently and correctly used and represented ○ Providing value to the solution
Record decisions, issues, and concerns.	
Identify potential fixes.	
Communicate issues and decisions to affected stakeholders.	Include the role of stakeholders.
Update the design to address identified issues.	Assign corrective action to the appropriate stakeholders.
Review the solution.	May include: • Conducting peer reviews of selected components • Performing verification of the components as appropriate
Revise the component as necessary.	

Example Work Products

Example Work Products	Further Explanation
Design evaluation issues	Identified deficiency or improvement with the design.
Design review meeting minutes	
Updated design	
Updated solution	

Related Practice Areas

Peer Reviews (PR)

Requirements Development and Management (RDM)

Verification and Validation (VV)

TS 2.3

Required Practice Information

Practice Statement

Provide guidance on use of the solution.

Value

Supports usability and maintainability of the solution.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Develop and provide guidance materials.	

Example Work Products

Example Work Products	Further Explanation
Guidance material	Guidance material may include: • Customer documentation • Operator or user instructions • Maintenance instructions • Installation instructions • Operations instructions • Administrator instructions • Online help • Training materials Ensure where appropriate that safety and security information is incorporated within guidance materials for using and maintaining the solution in all intended environments.

Level 3

TS 3.1

Required Practice Information

Practice Statement

Develop criteria for design decisions.

Value

Increases the likelihood of producing a robust design that meets customer requirements and constraints.

Additional Required Information

Record the name and detailed description for each criterion.

Explanatory Practice Information

Additional Explanatory Information

The criteria used to select solutions should provide a balanced approach to cost, requirements, benefits, and risks. Note that identified criteria may lead to the conclusion that no alternative solutions are necessary. Developing design criteria is often an iterative process. Design criteria may be different based on the domain and design activities.

Example Activities

Example Activities	Further Explanation
Analyze, develop, evaluate, use, and keep updated design criteria.	
Review and revise the design criteria with affected stakeholders as needed.	Revise criteria when requirements, budget, technology, or resources change or when a defect is discovered.

Example Work Products

Example Work Products	Further Explanation
Design criteria	Examples of criteria include: • Complexity • Cost of: ○ Development ○ Manufacturing ○ Procurement ○ Maintenance ○ Support

Example Work Products	Further Explanation
	• Time to implement • Technology availability and limitations • Resources needed • Performance • Robustness • Requirements and technology evolution • End user and operator capabilities • Rationale why alternative solutions are not required • Whether the solution needs to be: ○ Modular ○ Clear ○ Simple ○ Maintainable ○ Verifiable ○ Portable ○ Reliable ○ Accurate ○ Secure ○ Scalable ○ Usable

Context Specific

Data

Context Tag:	CMMI-DATA
Context:	Use processes to incorporate data management best practices as an integral part of the solution.

Address data needs, constraints, and requirements within design criteria. Data criteria should include considerations of governance, privacy, data security, regulations, access, and other controls.

Security

Context Tag:	CMMI-SEC
Context:	Use processes to incorporate security considerations as an integral part of the solution, work, project, and organization.

Address security needs, constraints, and requirements within design criteria. Design criteria should include previous known design flaws. Typical security design flaws may include lack of:

- Threat modeling
- Secure design patterns
- Secure design principles
- Reference architecture

- System, operational, and data administration and management, e.g., Single Sign-On (SSO), Multi-factor Authentication (MFA)

TS 3.2

Required Practice Information

Practice Statement

Develop alternative solutions for selected components.

Value

Ensures that the most beneficial solution is identified and selected.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Any feasible solution that aligns with the customer's requirements can be an alternative solution.

Alternative solutions can include developing new technologies or identifying and applying current technologies in different ways.

Solutions can be based on past designs. Consider alternatives that address and perform the same necessary functions in different ways.

Example Activities

Example Activities	Further Explanation
Develop, identify, and record alternative solutions.	May include: • New and currently in use product technologies • Reusable solutions or solution components • Externally-provided solutions ○ Commercially available ○ Customer provided ○ Publicly or community available • Functionality and quality attributes • Terms and conditions of warranties for the products
Record the requirements allocation for each alternative.	
Communicate results.	

Example Work Products

Example Work Products	Further Explanation
Alternative solutions	

TS 3.3

Required Practice Information

Practice Statement

Perform a build, buy, or reuse analysis.

Value

Ensures that the most effective way to implement the design has been chosen.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

A “build, buy, or reuse analysis” or a “trade study” is typically used to ensure that the most beneficial, technically viable option is selected to implement the solution. It is based on an analysis of the needs of the project. Start the analysis early in the project, continue it during the design process, and complete it with a decision on how to proceed.

Example Activities

Example Activities	Further Explanation
Perform a build, buy, or reuse analysis.	A decision analysis and resolution method or approach may be used to address the design criteria and provide the rationale for making the decision.
Record analysis and communicate results.	

Example Work Products

Example Work Products	Further Explanation
Build, buy, or reuse analysis	Factors affecting a build, buy, or reuse analysis can include: • Functions the solutions will provide • Available project resources and skills • Costs of acquiring versus developing internally • Critical delivery and integration dates • Market research of available products • Functionality and quality of available solutions

Example Work Products	Further Explanation
	• Skills and capabilities of potential suppliers • Licenses, warranties, responsibilities, and limitations associated with solutions being acquired • Availability • Proprietary issues • Regulatory or legal issues • Risk reduction

Related Practice Areas

Decision Analysis and Resolution (DAR)

TS 3.4

Required Practice Information

Practice Statement

Select solutions based on design criteria.

Value

Ensures the most efficient and effective solution is selected to meet the customer's requirements within cost, schedule, and performance constraints.

Additional Required Information

Record the rationale for selecting the solution.

Explanatory Practice Information

Additional Explanatory Information

Record solution descriptions and rationale for selection or rejection. Update the record throughout development as solutions and detailed designs are developed and implemented. Maintain a record of rationale to aid future decision-making. Records prevent rework, rehashing decisions, and provide insights for applying new technology as it becomes available.

Example Activities

Example Activities	Further Explanation
Evaluate each alternative solution against the selection criteria.	Identify and resolve issues with the alternative solutions and requirements. High-risk situations may use simulations, prototypes, or pilots to assist in the evaluation.
Select the solutions that satisfy the established criteria.	

Example Activities	Further Explanation
Based on evaluation of alternatives, reassess and update selection criteria when necessary.	Update criteria if additional derived requirements or criteria are discovered.
Develop, use, and keep updated records of solutions, evaluations, and rationale.	

Example Work Products

Example Work Products	Further Explanation
Recorded solutions, evaluations, and rationale	Includes rationale for selection or rejection.

Related Practice Areas

Decision Analysis and Resolution (DAR)

TS 3.5

Required Practice Information

Practice Statement

Develop, keep updated, and use information needed to implement the design.

Value

Avoids rework by ensuring that solution implementers have the information they need to develop a solution that meets the customer's requirements.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

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Example Activities

Example Activities	Further Explanation
Record the information needed to implement the solution.	The technical description defines the required design configuration and procedures to ensure the performance of the item is adequate. Provide a technical description of the solution that addresses such aspects as: • Development • Production • Logistics • Maintenance

Example Activities	Further Explanation
	• Operations and support • Solution lifecycle states and changes
Revise the information needed to implement the solution, as needed.	Revise information as scope, requirements, or criteria change.

Example Work Products

Example Work Products	Further Explanation
Technical data package	Commonly used to implement the design when the implementation is complex or broken into multiple parts, e.g., subsystems, internal or external team or supplier. Package information to implement the design may include: • Product definition data • Engineering drawings • Specifications • Standards • Performance requirements • Reliability data • Packaging details • Modeling data • Version control information • Verification and validation criteria • Any other information needed by the solution implementers
Requirements, design, test, and traceability information	

TS 3.6

Required Practice Information

Practice Statement

Design solution interfaces or connections using established criteria.

Value

Reduces the likelihood of failures and rework during testing and operations and maximizes performance.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

This section left blank for future content.

Example Activities

Example Activities	Further Explanation
Define interface or connection criteria.	When defining criteria for interfaces or connections, define or investigate critical parameters to determine if they are applicable. These parameters are often specific to a given type of system and are often associated with quality attribute requirements, e.g., safety, security, durability, and mission critical characteristics.
Develop interface or connection design alternatives using established criteria.	
Identify interfaces or connections, both internal and external.	
Identify interfaces or connections between components and related processes.	For example: • Interfaces or connections between fabricated components and the fixtures used during the manufacturing process • Interfaces or connections to test beds for software testing
Identify user interfaces or connections.	Examples of users include: • Developers • Testers • Operators • Users Each may have their own needs, norms, and perspectives that may influence how they interact with the solution.
Record, keep updated, use, and communicate selected interface or connection criteria, design, and selection rationale.	

Example Work Products

Example Work Products	Further Explanation
Interface or connection specification criteria	May include: • Safety

Example Work Products	Further Explanation
	• Security • Performance • Standards • Capacity
Interface or connection design specification	Consistently and correctly designed interfaces or connections, can include: • Origin • Destination • Sequencing constraints or protocols • Resources needed or consumed • Exception or error handling behavior for inputs that are erroneous or out of specified limits • Electrical, mechanical, and functional characteristics • Interfaces or connections between solutions • Interfaces or connections with users, operators, and maintainers of the solution
Interface or connection control documents	
Rationale for selected interface or connection design	

